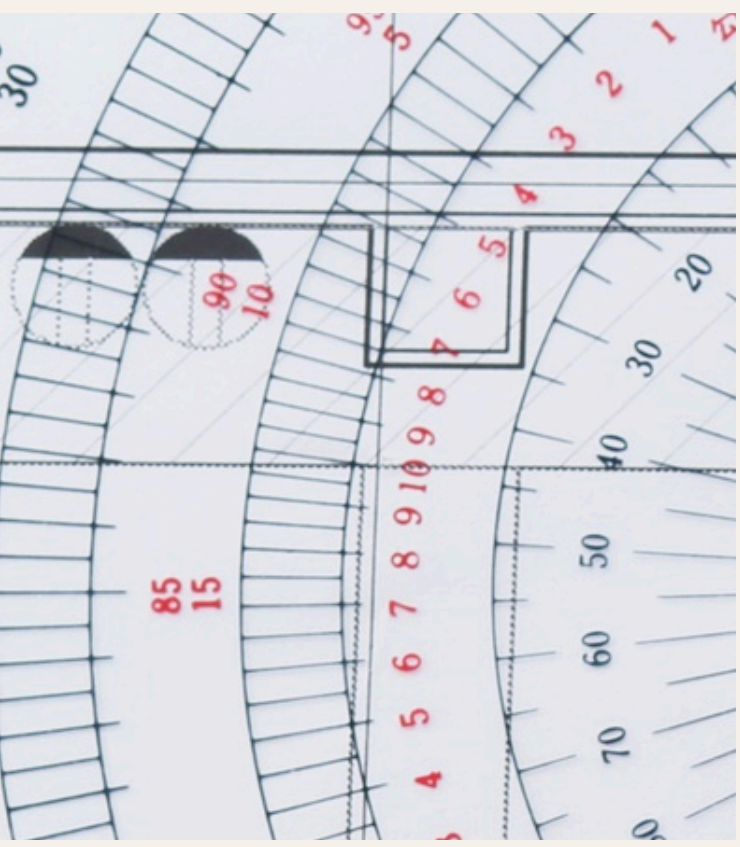
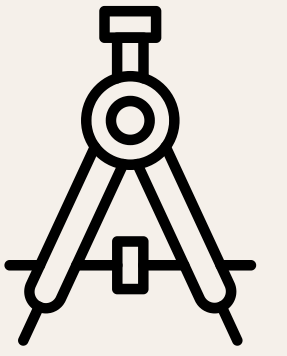


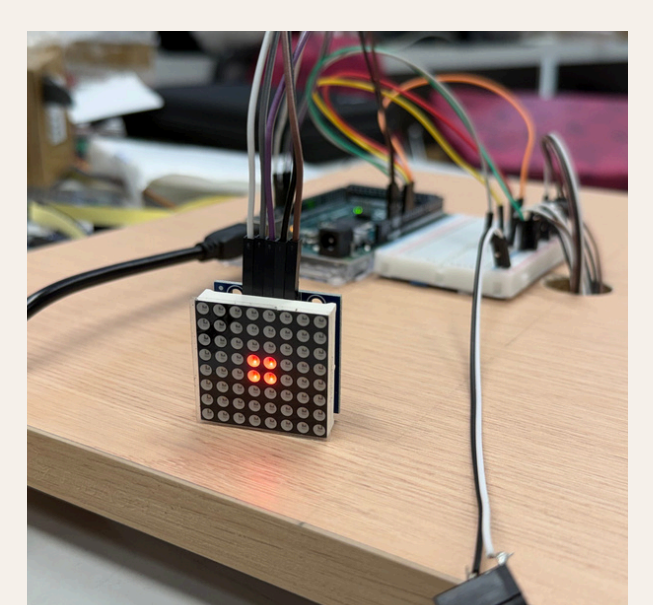
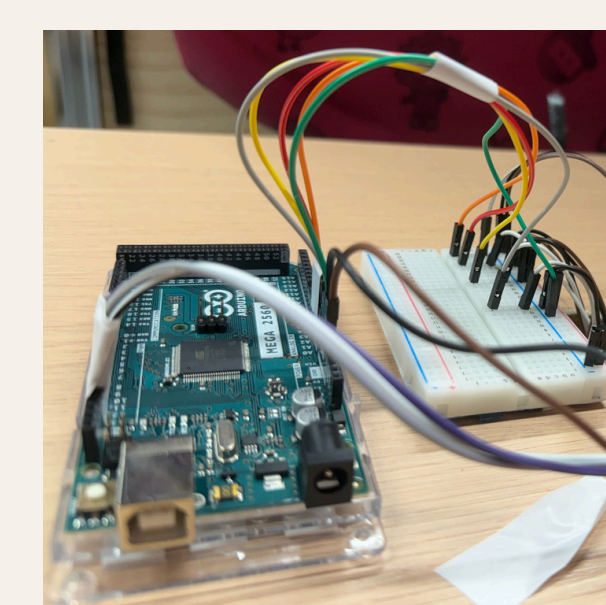


BME31147 BME Innovation for the Community (Service-learning)

UPCYCLED SMART SEATING SYSTEM FOR ELDERLY PEOPLE



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Project Background

- **Core Purpose:** Designed for visually impaired elderly users to support safe posture, independent mobility, and comfort.
- **Key Requirements:** Emphasizes safe sit-to-stand movement, pressure-care awareness, and fall prevention.
- **Collaboration & Design:** Developed via a strategic partnership with Kowloon Motor Bus (KMB) using upcycled transport materials, guided by expert occupational therapy insights.

End Users Needs

1. Chair should be not too heavy
2. Should be stable, not collapse
3. User-friendly
4. Use of durable material
5. Prioritize comfort of users
6. Flexibility to incline the chair
7. Integration of sensors
8. Include seat belt to stabilize user

Methodologies & Technologies

Hardware

- **Upcycled Structural Design:** Partnered with Kowloon Motor Bus (KMB) to repurpose durable transport-seating materials, creating a sturdy, modular aluminum frame featuring an anti-topple base.
- **Ergonomic Adjustability:** Incorporated user-centric physical supports, including adjustable armrests, multi-positional backrest inclinations

Software

- **IoT Sensor Integration:** Embedded load cells and strain gauges strategically within the seat to monitor presence, track prolonged sitting duration, and measure real-time Centre of Pressure (COP) shifts.
- **Real-Time Feedback:** Linked sensor data to an intuitive LED matrix display, providing actionable insights for posture correction, pressure-care management, and fall prevention.

Testing Results & User Feedback

Strengths

- Stable and safe structure
- Easy to clean surface
- Strong and durable build
- Adjustable backrest adds flexibility
- Suitable for larger users

Weaknesses

- Chair too wide for average user
- Seat height too high for average user
- Cushion too hard for long sitting
- Insufficient lower back support
- Headrest needs to be more stable
- Too heavy for moving around

Limitations and Future Work

Hardware Improvements

- Lower the armrests
- Reduce overall chair width/size
- Add seat height adjustment
- Use a softer/thicker cushion
- Improve lumbar support
- Refine headrest design
- Use of seatbelt to secure the user

Software Improvements:

- Display Line-of-Sight Blockage of LED matrix
- Lack of Averaged Data Logging
- Advanced Pressure Mapping Integration on seating and backrest
- Integration of Alert System for independent repositioning by the elderly